Sanjukta Das

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WORK EXPERIENCE

Business Intelligence & Analyst Intern
Intel Corporation, AZ June - December 2019
- Collaborated with multiple teams – Operating Margin Task Force, IOTG-Finance, Sales & Marketing Group(SMG) – for data collection, analysis, modeling & visualization.
- Performed
  - Exploratory Data Analysis(EDA) of financial, and administrative data contained within the organization's data warehouse to maximize insight, detect underlying structure, important variables, outliers & anomalies, test underlying assumptions, etc.
  - Statistical analysis like hypothesis testing for validation and verification of data models.
- Designed, developed and maintained key indicators necessary to deliver analytical findings.
- Developed and executed strategy for gathering and analyzing data, investigate data trends, anomalies and business opportunities.
- Designed & developed
  - Data visualizations and real-time dashboards to drive standardization and year-round tracking of IOTG Operating Margin.
  - Executive view dashboard with SKU level margin tool for Sales & Marketing Group team to set margin goals and allow comparable margin visibility.
  - Interactive team website with one stop project management repository using SharePoint, HTML, CSS and JavaScript.
- Extensive use of Power BI to produce operational dashboards and visualization in support of in-depth analysis.
- Achieved an accuracy of more than 85% on the data model(OLAP Cube) built for the margin tool embedded in the executive view dashboard for SMG team.

HR-Executive – US Payroll
HSBC HDPI, India February 2014 - 2015
- Part of HR operations payroll for internal HSBC employees in USA.
- Handling New Hires, Contingent workers, Personal Changes, Approvals, Employee Self Service, Manager Self Service Transactions, pay changes, On Call Pay queues, resolve queries and escalation cases, promotions, demotions, employee verifications.
- Administer onboarding, verifying post hiring procedures, create & update Record of Employment (ROE), payroll format and Taxation.
- Maintenance of database, including high to low profile personnel in Group HR Systems (People Soft portal), produce daily & monthly Taleo reports, biweekly audit reports.

Business Development Analyst – Modern Trade
Parle Agro, India January 2013 - 2014
- Administering both primary and secondary revenue data of Modern Trade division.
- Responsible for producing daily, weekly & monthly basis revenue report, PO Tracking & updating MBQ of the products.
- Oversee channeling of stock from manufacturer through the intermediaries to the consumer.
- Develop and execute promotion strategy.

Finance Analyst Intern
IDBI Capital, India April – September 2012
- Study factors affecting stock market volatility & investor perception towards market trading.
- Data collections, analysis, and investigate trends in stock market data of 5 years i.e., from 2007 to 2011.
- In these sessions, stock market was most volatile so I covered various analyses with most affected factors to the global market.

Finance Analyst Intern
OHPC, India May – July 2011
- Study Profitability and Performance of OHPC Limited for three financial years.
- prime role was to perform financial analysis, further more involved in the measuring the various ratio analysis and explicitly focusing on the working capital.
- Collect accurate picture of strength and stability of the organization, know the Operational efficiency, to find out the profit and loss made by the business at the end of the interval and ascertain the financial position.
- Synthesizing and analyzing financial data (like budgets and income statement forecasts) with consideration for the company's goals.
- Generate reports & visualization of financial information, and prepare a comparative study for 3 consecutive years on the performance & financial attributes of the organization.
- Recommend alternate courses of actions to reduce costs and improve the company's finances.
- Suggest methodology for improvement in profitability and liquidity position.

EDUCATION

M.S. in Computer Science (3.55/4)
Portland State University Spring 2019 – Present

Post-bacc in Computer Science (3.33/4)
Portland State University 2017 – 2018

Computer Science (Transfer)
Portland Community College 2016 – 2018

MBA in Finance
Regional College of Management, India 2011 – 2013

SKILLS

Technical:
Python, R, RMarkdown, C, C++, Java, Data Structure, Algorithm, HTML, CSS, Javascript, ReactJS, Bootstrap, SharePoint, Wireframing, Power BI, Tableau, Linux, Flask, Amazon Web Services (AWS), Google Cloud Platform (GCP), Azure, Docker, Kubernetes, Github, Bitbucket, MongoDB, SQL, PostgreSQL, SDLC, Object-Oriented Programming, Data Mining, Data Wrangling, Data Modelling, Data Analysis, EDA, JMP, Structural Analysis, Product Roadmapping, JUnit Testing, JSP, Servlet, REST API, Google Web Toolkit, Android, C#, .NET, OLAP, Microsoft Office, Excel, VBA

Business:
Ratio Analysis, Cash Flow Management, Security Analysis, Capital Budgeting, Trading & P/L, Balance Sheet Reports, Financial Reporting.

COURSE WORK

Technical:
Data Visualization, Machine Learning, Algorithm, Internetworking Protocol, DBMS Implementations, Computer Vision, Frontend Development, Full-Stack Web Development, Internet and cloud Systems, Cloud & Cluster database, Advanced Java Programming, Data Structure, Operating Systems, Linux, Software Engineering, Computational Photography, Database Management System, Engineering & Technology Roadmapping etc.

Business:
Economics, Financial Analysis, Capital Budgeting, Business Statistics, Research Methodology, Insurance & Risk Management, Quantitative Techniques, Financial Accounting, Financial Management, Economic Environment of Business, Management Information System, Working Capital Management, Management Accounting for Decision Making, Operations Management, Banking & Insurance Management, Financial Services, Security Analysis & Portfolio Management, International Finance, Financial Planning and Wealth Management, Financial Derivative, Business Valuation & Financial Modelling, Project Planning & Control, etc.

ACADEMIC PROJECTS

Deep Learning (Python):

- **Auto Tagging Stack Overflow Questions –**
  - Implement a tag suggestion system using both Machine Learning and Deep Learning models. Performance evaluation of both the approaches.
  - Implemented classifiers - SGC Classifier, Multinomial Naïve Bayes Classifier, Random Forest Classifier, etc., using pandas, numpy, matplotlib, seaborn, pickle, sklearn, bs4 (beautiful soup 4), nltk.tokenize, logging, scipy.sparse and Stack sample dataset (10% of stack overflow Questions and answers)

Machine Learning (Python):

- **Stock Price Prediction –** implementation & performance evaluation of three Machine Learning models (i.e. Linear Regression, SVR (Support Vector Regression), LSTM (Long Short Term Memory)) for forecasting the future stock prices. It also involves validation of the models using real datasets acquired from Google Finance, Kaggle and Yahoo Finance; and performance and accuracy evaluation of the models by comparing and analyzing it against some typical baselines.
- **Clustering –** Implemented K-Means and Fuzzy Means, unsupervised machine learning algorithms using Numpy and Matplotlib libraries, and a dataset consisting 1500 data points in 2-dimension from 3 Gaussians, with considerable overlap.
- **Spambase –** This project involves solving a classification problem in ML i.e. identifying spam/non-spam emails from Spambase datasets using linear SVMs
- **Digit Classification –** Recognized handwritten digits from MNIST Dataset by implementing perceptron learning algorithm. Trained 10 perceptron to learn and classify as a group, the handwritten digits in the MINST dataset, where each perceptron has 785 inputs consisting of 784 pixels i.e., 28x28 pixel image represented as gray scaled value (0 – 255) and one output i.e., either 0 or 1. Each perceptron's target is one of the 10-digits (0-9) and it learns using the perceptron learning algorithm.

Data Visualization (R): Multiple data visualization projects on Museum of Modern Arts(MOMA), COVID-19, ArcGIS Maps, developing static and dynamic website using Shiny, Leaflet, ArcGIS, Tidyverse, ggplot, dplyr, RMarkdown, blogdown, Hugo, etc.

Database:

- **PostgreSQL Benchmarking(Python & SQL) –** A multi-phased project based on Wisconsin benchmarking, aspired to evaluate PostgreSQL database with assorted configuration parameter settings and optimizer options; that implicates designing and implementation of various experiments for the performance benchmarking. It involved generation and storage of variable sized (100K to 100M tuples) datasets of strings, designing and implementation of queries, working with PostgreSQL's configuration settings, index, query optimization, joins, scans while executing the queries.
- **MongoDB(Java) –** Hands-on experience of data preprocessing & wrangling, designing schema, data modelling, storage and application design with cloud-based data management system in MongoDB Atlas, using Freeway dataset along with designing and implementation various queries in Java as well as MongoDB query language.

Computer Vision (Python & C++):

- **Video Frame Interpolation –** developed a tool that converts a video with 24 or 30 fps to 48 or 60 fps, by interpolating one frame between any two consecutive frames in the original low fps video. Basically, estimate the optical flow between two consecutive frames and generate the middle frame guided by the flow.
- **Panorama –** Developing a program to stitch multiple images to obtain a wide seamless panorama, using Scale Invariant Feature Transform (SIFT) to extract local features in the input images, implemented K-Nearest Neighbors algorithm and Random Sample Consensus (RANSAC) to match the features and compute the homographic matrix for image wrapping respectively and followed by application of weighted matrix as a mask for image blending.
- **Color Transfer –** Implemented a program that transfers the color characteristics of one image to another.
- **Object removal –** implemented a program inspired from resynthesizer and pattern matching for removing object in an image, using OpenCV and Numpy library and Inpaint for better blending.
- **Toonify –** gives a cartoon like effect to images. Emulated cel-shading effects by applying median filter, canny edge detection, morphological operation, edge filtering and bilateral filter.
- **Mosaic –** gives a tilling effect, implemented by splitting the image to tiles, then averaging the color values and replacing each tile with a matching image.
- **Circle Packing –** gives the effect of cluster of circles by storing all the pixel values to fill the circles and compare the (x, y) position within the image bounds to prevent overlapping of circles.

Frontend Web Development:

- **Marvel Comics Browser(React JS) –** Developed a user-friendly and interactive website using ReactJS to explore comics by series & super heroes of Marvel Universe.

Full-Stack Web Development:

- **Instapix (Flask, SQL, gcloud storage, Cloud Vision & Translate API) –** Adopted MVC (Model-View-Controller) architectural pattern to design and develop a Google cloud-based social networking application using Flask web framework of Python, that allow users to follow other users, share/comment/like posts/photos, and view timeline etc.
- **Bubble Tea (Flask, SQL, GCP) –** Designed and developed a Google cloud-based application to allow users to add different Bubble Tea store details (name, location, hours, contact, drinks, etc.) and give reviews.

Networking:

- **Internet Relay Chat(Java) –** Designed & implemented an Internet Relay Chat (IRC) application layer protocol for real-time text-based messaging in Java. It provides a generic message passing framework for multiple clients to communicate with each other via a central forwarding server.

Software Applications:

- **Web Crawler (Python, BS4) –** traverse the world wide web to create search engine index with given seed links.
- **Chat Application (Java) –** A Text-based messaging application using Java swing for designing GUI, HTML for aligned display of messages, Java socket for client-server communication, MongoDB to persist user's information.
- **Phone Bill (Java, GWT) –** web application that maintains the phone call record details. Features implemented are: display a README manual describing how the program works, adding a new call record, display details of all calls or calls recorded within a certain period of time.
- **Twitter (Java) –** An application which emulates the Twitter. Given a set of tweets as input, it extracts information such as date and time, user, tags, keywords, etc. and categorizes the tweets based on it.

Data Structures & Algorithm (Java & C++):

- **String Matching –** Given a text, find the occurrences of a pattern in it and return the locations of match. Implemented Brute force, Boyer Moore, KMP, Shift-OR, Rabin-Karp, in C++ to performance benchmark multiple string-matching algorithms given randomly generated patterns & Homo_sapiens DNA sequence datasets of variable sizes, to find the one with the best and the worst-case run-time and space complexity.
- **Sorting Algorithms –** performance evaluation of various sorting algorithms using randomly generated datasets.
- **Graph Algorithm:** implemented Kruskal's and Prim's method of finding the Minimum Spanning Tree on random graphs, using union find data structure.
- **Shunting Yard –** Dijkstra's shunting yard algorithm using stack to evaluate logical expressions with AND, OR, NOT, etc. based on their order of precedence.
- **Multiply linked list –** implemented linked list with three separate link chains to keep the data entries unique and sorted by three different fields.
- **AVL Tree –** set container like in STL using AVL tree that supports union, intersection, cross product, and power set operations.

Operating Systems:

- **XV6 (C, UNIX) –** Implemented **System calls:** a new system calls for date and time. **Process:** system calls to support and obtain process ownership, set and extract UID, GID, PID, track amount of time a process uses CPU. **Shell:** create new user command ('ps' and 'ctrl+p' command), extract process detailed information. Scheduling: implemented MLFQ scheduler. **File system:** chmod, chown, chgrp system calls to set and modify file permissions that persist across xv6 sessions. Modified 'exec' and 'ls' command to enforce and display new file permissions respectively.
- **Buddy System (C) –** module that imitate malloc and free to manage a block of memory (e.g. heap)

Android:

- **Hangman (Android) –** Developed a word guessing game given a limited number of wrong guesses. It selects and sets a random word from a text file and asks the user to guess the letters. Displays letters or next hanging picture on every correct or wrong guess respectively. The user loses once the hangman picture is complete.

REFERENCES

Available on request.